Algorithmic and Minimax Complexities in Kernel Bandits

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

Abstract

Gaussian-process upper confidence bound (GP-UCB) and decision-estimation-coefficient (DEC) methods may appear, at first sight, to belong to different theories. This paper places the two viewpoints in a common algorithmic-information language for frequentist RKHS bandits. GP-UCB fixes an algorithmic, rather than true, Gaussian-process prior and exploits realized-trajectory complexity together with computational tractability, whereas MAMS optimizes a robust class-wide MAIR/DEC envelope. Through the unified MAIR framework and heterogeneous positive-semidefinite algorithmic priors, we generalize both the GP-UCB analysis and the MAMS algorithm, propose a safeguarded master that combines their advantages, and provide a kernel-bandit construction showing that algorithmic complexity can be more informative than class-wide minimax or DEC certificates in overparameterized models. The resulting message is that algorithmic information and class-wide minimax coefficients answer different questions and can lead to different gaps; kernel bandits provide a clean setting in which this distinction becomes mathematically visible.

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1 Introduction

Kernal and Gaussian-process bandits are a useful meeting point for two cultures of sequential learning. The first culture is algorithmic and practical. GP–UCB maintains a kernel ridge-regression posterior mean and variance, then queries the point with the largest upper confidence bound (Srinivas et al., 2010). GP-UCB and its variants have became influential because they are simple, reliable, and easy to implement in AI and decision-making applications. The second culture is information-theoretic and minimax. The DMSO framework and the decision-estimation coefficient (DEC) characterize interactive learning through an offset between decision regret and statistical information (Foster et al., 2021, 2023; Chen et al., 2024). Algorithms such as EBO, E2D, and MAMS solve minimax versions of this offset (Xu and Zeevi, 2025).

The standard frequentist kernel-bandit model is not literally Bayesian. The truth is a fixed function $f^\star$ in an RKHS and the observation noise is conditionally sub-Gaussian. The Gaussian process used by GP–UCB is an algorithmic reference: it defines a posterior mean, a posterior variance, and a log-determinant information budget. This is why we use the phrase *kernel/Gaussian-process bandits*. The statistical model is best called a kernel or RKHS bandit; the algorithmic reference is Gaussian-process shaped. The central claim of the paper is that GP-UCB and DEC are two projections of a unified MAIR identity, with kernel bandits providing a clean setting in which the distinction between algorithmic information and class-wide regret-information coefficient can be made mathematically precise.

This perspective helps clarify the long discussion around the information-theoretical optimality of GP–UCB. The original GP–UCB analysis of Srinivas et al. (2010) bounds regret by maximal information gain. Later work sharpened information-gain estimates and raised the problem of tight online confidence intervals for RKHS elements (Vakili et al., 2021b,a). Whitehouse et al. (2023) showed that improved smoothness-aware regularization and Hilbert-space self-normalized concentration yield nearly optimal, and in particular sublinear, GP–UCB rates for important kernels such as Matérn kernels. In the Bayesian GP setting, Iwazaki (2025); Takeno and Iwazaki (2026) further emphasize that the realized information gained by the GP–UCB trajectory can be much smaller than the maximum-variance-reduction trajectory. These developments all point to the same lesson: the realized algorithmic information and the regret-information coefficient should be kept separate from the full class-wide worst case. On the other hand, Lattimore (2023) gives lower-bound evidence that fully general adaptive confidence intervals cannot be arbitrarily tightened, and Wang and Zhang (2026) gives a Matérn lower-bound obstruction for GP-UCB under polynomial effective optimism.

Starting from finite model classes, DEC (Foster et al., 2021) and MAIR (Xu and Zeevi, 2025) aim to design robust class-wide algorithms for full-class minimax regret problems through regret–information coefficients. In this setting, DEC provides matching lower-bound certificates on the regret–information coefficient Foster et al. (2021, 2023). However, their finite-covering implementation can be pessimistic for kernel problems: discretization and covering costs may dominate the clean log-determinant estimation complexity that makes GP–UCB practical. Conversely, GP–UCB is not automatically justified on the MAIR/DEC coefficient side: optimism controls the coefficient through confidence widths, but does not solve the minimax offset problem. The unifying view is therefore not "GP–UCB beats DEC" or "DEC replaces GP–UCB." It is that algorithmic performance, Bayesian-average performance, and class-wide minimax performance are distinct and should be compared explicitly.

Contributions. The paper makes three contributions.

1. We formulate GP–UCB and MAMS/DEC in a common MAIR language. GP–UCB is viewed as a calibrated algorithmic-prior method: a surrogate GP posterior supplies the information budget, while self-normalized calibration and optimism verify the MAIR gradient-error condition. MAMS is viewed as the robust saddle-point counterpart: it optimizes a class-wide

envelope to give direct DEC/MAIR control. This gives a direct comparison between realized algorithmic information and class-wide minimax coefficients in kernel bandits.

2. We develop two complementary extensions of this comparison. On the GP–UCB side, we allow heterogeneous positive-semidefinite algorithmic priors, so that the posterior geometry and the information gain are chosen by the algorithm rather than fixed by a scalar ridge. On the MAMS side, we pass from finite model classes to RKHS/kernel classes through finite covers and discretization error. We also propose a safeguarded master that combines a practical heterogeneous GP–UCB base algorithm with a robust MAMS-type benchmark.

3. We give a heterogeneous-prior hub–cloud construction separating algorithmic and class-wide minimax complexity. The example shows that full-class minimax or DEC certificates can be pessimistic for the realized performance of a regularized algorithm, especially in overparameterized kernel-bandit models. It therefore supports a more refined view of the GP–UCB optimality debate: minimax guarantees remain important, but algorithmic information and the chosen posterior geometry are also essential performance measures.

2 Kernel bandits with heterogeneous algorithmic priors

2.1 DMSO notation and regret

We use standard DMSO notation. A model $M \in \mathcal{M}$ specifies, for every decision $\pi \in \Pi$, an expected reward $r_M(\pi)$ and an observation law $P_{M,\pi}$. Let

$$\pi_M \in \operatorname*{argmax}_{\pi \in \Pi} r_M(\pi)$$

be an optimal decision under M. For a randomized decision $p \in \Delta(\Pi)$ define the one-step regret gap

$$\Delta_M(p) := r_M(\pi_M) - \mathbb{E}_{\pi \sim p} r_M(\pi).$$

For an algorithm ALG and a fixed ground truth $M^\star$, the expected cumulative regret is

$$\mathfrak{R}_T(\mathsf{ALG}, M^\star) := \mathbb{E}_{M^\star,\mathsf{ALG}}\left[\sum_{t=1}^{T} \{r_{M^\star}(\pi_{M^\star}) - r_{M^\star}(\pi_t)\}\right]. \tag{1}$$

We also use the Bayesian and minimax risks

$$\mathfrak{R}_T^{\mathrm{Bayes}}(\mathsf{ALG}, \Pi_0) := \mathbb{E}_{M \sim \Pi_0} \mathfrak{R}_T(\mathsf{ALG}, M), \qquad \mathfrak{R}_T^{\mathrm{mm}}(\mathcal{M}) := \inf_{\mathsf{ALG}} \sup_{M \in \mathcal{M}} \mathfrak{R}_T(\mathsf{ALG}, M).$$

The core thesis is that these three risks can have different scales for the same ambient kernel class.

2.2 Kernel bandits and matrix-valued ridge geometry

Let $\mathcal{X}$ be a finite action set and let $\phi(x) \in \mathbb{R}^d$ be a feature map. A frequentist linear/kernel truth is

$$f^\star(x) = \langle \theta^\star, \phi(x) \rangle,$$

and observations satisfy

$$Y_t = f^\star(x_t) + \varepsilon_t, \tag{2}$$

where x_t is H_{t-1}-measurable and ε_t is conditionally R-sub-Gaussian. Infinite-dimensional RKHS statements follow by the usual limiting argument; the finite-dimensional notation keeps the role of the matrix regularizer transparent.

Fix a self-adjoint positive-semidefinite precision matrix $\Gamma \succeq 0$ and a reference noise level $\lambda > 0$. To avoid domain distractions, the displayed finite-dimensional formulas are written for $\Gamma \succ 0$. The genuinely semidefinite case is obtained either by using Moore–Penrose inverses on $\mathrm{range}(\Gamma)$ or by replacing Γ with $\Gamma + \zeta I$ and sending $\zeta \downarrow 0$ whenever the resulting code length and information gain are finite. The corresponding algorithmic Gaussian reference is

$$\theta \sim N(0, \lambda\Gamma^{-1}), \qquad Y_t \mid \theta, x_t \sim N(\langle \theta, \phi(x_t)\rangle, \lambda). \tag{3}$$

Equivalently, the algorithmic covariance kernel is

$$k_\Gamma(x, x') := \lambda \phi(x)^\top \Gamma^{-1} \phi(x').$$

Only the prior covariance $\lambda\Gamma^{-1}$ and the observation variance λ enter the posterior formulas. If one wants to regard the prior covariance C as fixed independently of the observation noise, set $\Gamma = \lambda \mathsf{C}^{-1}$; then $k_\Gamma(x, x') = \phi(x)^\top \mathsf{C} \phi(x')$. The standard GP–UCB geometry is the scalar special case. A heterogeneous prior corresponds to a non-scalar Γ: large precision means that a direction is algorithmically expensive; small precision means that the direction is favored by the prior. Thus the scalar ridge parameter is not separate from the prior geometry; it is the isotropic instance of the general PSD precision matrix.

Define

$$V_t^\Gamma := \Gamma + \sum_{s=1}^t \phi(x_s)\phi(x_s)^\top, \qquad S_t := \sum_{s=1}^t \varepsilon_s \phi(x_s).$$

The algorithmic posterior mean and latent variance are

$$\mu_t^\Gamma(x) = \phi(x)^\top (V_t^\Gamma)^{-1} \sum_{s=1}^t Y_s \phi(x_s), \qquad \sigma_{t,\Gamma}^2(x) = \lambda \phi(x)^\top (V_t^\Gamma)^{-1} \phi(x). \tag{4}$$

The realized algorithmic information gain is

$$\mathcal{I}_{T,\Gamma} := \frac{1}{2} \log\det\left(I + \Gamma^{-1/2} \Big(\sum_{t=1}^T \phi(x_t)\phi(x_t)^\top \Big) \Gamma^{-1/2} \right) = \frac{1}{2} \log\det(I + \lambda^{-1} K_{\Gamma,T}), \tag{5}$$

where $K_{\Gamma,T} = (k_\Gamma(x_i, x_j))_{i,j \le T}$. The one-step increment is

$$g_t^\Gamma := \frac{1}{2} \log\left(1 + \lambda^{-1} \sigma_{t-1,\Gamma}^2(x_t)\right), \qquad \sum_{t=1}^T g_t^\Gamma = \mathcal{I}_{T,\Gamma}. \tag{6}$$

2.3 Posterior-reference information and fixed-truth transfer

Let q_1 be a reference prior over models, q_t the Bayesian posterior after H_{t-1} under a reference observation channel, and

$$Q_t^\pi(\cdot) := \int P_{M,\pi}(\cdot) q_t(dM)$$

the posterior predictive law. The following identity is the bridge from finite MAIR potentials to GP information gain.

Theorem 2.1 (Posterior-reference density-ratio telescope)**.** *Assume the likelihood ratios below exist. For predictable decisions π_t,*

$$\frac{dq_{T+1}}{dq_1}(M) = \prod_{t=1}^T \frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t). \tag{7}$$

Consequently,

$$\log \frac{dq_{T+1}}{dq_1}(M) = \sum_{t=1}^{T} \log \frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t).$$

Under the Bayesian joint law $M \sim q_1$ and $O_t \sim P_{M,\pi_t}$,

$$\mathbb{E} \log \frac{dq_{T+1}}{dq_1}(M) = \sum_{t=1}^{T} \mathbb{E} I_{q_t}(M; O_t \mid H_{t-1}, \pi_t). \tag{8}$$

Proof. Bayes' rule gives

$$\frac{dq_{t+1}}{dq_t}(m) = \frac{dP_{m,\pi_t}}{dQ_t^{\pi_t}}(O_t).$$

Multiplying the Radon–Nikodym derivatives proves (7). Taking logarithms and conditional expectation under $M \sim q_t$ gives

$$\mathbb{E}\left[\log \frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t) \mid H_{t-1}\right] = I_{q_t}(M; O_t \mid H_{t-1}, \pi_t),$$

and summing proves (8). □

Corollary 2.2 (Gaussian evaluation: log determinant)**.** *For the heterogeneous Gaussian reference* (3)*,*

$$I_{q_t^\Gamma}(\theta; Y_t \mid H_{t-1}, x_t) = \frac{1}{2} \log \left(1 + \lambda^{-1} \sigma_{t-1,\Gamma}^2(x_t)\right),$$

and hence

$$\sum_{t=1}^{T} I_{q_t^\Gamma}(\theta; Y_t \mid H_{t-1}, x_t) = \mathcal{I}_{T,\Gamma}.$$

Proof. Conditionally on H_{t-1}, the reference posterior gives $f(x_t)$ Gaussian with variance $\sigma_{t-1,\Gamma}^2(x_t)$, while Y_t has predictive variance $\lambda + \sigma_{t-1,\Gamma}^2(x_t)$. Subtracting Gaussian differential entropies gives the one-step formula. The determinant identity follows from the matrix determinant lemma applied sequentially to the posterior variance update. □

The fixed frequentist expectation is different. Evaluating the posterior ratio at a fixed truth gives an information-density term, not mutual information. The next lemma gives the exact transfer bound needed for frequentist GP–UCB.

Lemma 2.3 (Fixed-truth evaluation under a heterogeneous prior)**.** *Let $\theta^\star$ be fixed, let $Y_T = (Y_1, \dots, Y_T)^\top$, and let Φ_T be the $T \times d$ matrix with rows $\phi(x_t)^\top$. Set $f_T^\star = \Phi_T \theta^\star$ and $K_{\Gamma,T} = \lambda \Phi_T \Gamma^{-1} \Phi_T^\top$. For the Gaussian reference* (3)*, using the canonical likelihood-ratio version of the Radon–Nikodym derivative,*

$$\log \frac{dq_{T+1}^\Gamma}{dq_1^\Gamma}(\theta^\star) = \mathcal{I}_{T,\Gamma} + \frac{1}{2} Y_T^\top (K_{\Gamma,T} + \lambda I_T)^{-1} Y_T - \frac{1}{2\lambda} \|Y_T - f_T^\star\|_2^2. \tag{9}$$

Consequently,

$$\log \frac{dq_{T+1}^\Gamma}{dq_1^\Gamma}(\theta^\star) \leq \mathcal{I}_{T,\Gamma} + \frac{1}{2\lambda} (\theta^\star)^\top \Gamma \theta^\star. \tag{10}$$

The same inequality holds after expectation under any frequentist law for the noise and the algorithmic randomization.

Proof. For the realized design, the reference marginal law of Y_T is $N(0, K_{\Gamma,T} + \lambda I_T)$, whereas the reference likelihood at $\theta^\star$ is $N(f_T^\star, \lambda I_T)$. Taking the log likelihood ratio gives (9). The variational identity

$$Y_T^\top (K_{\Gamma,T} + \lambda I_T)^{-1} Y_T = \inf_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{\lambda}\theta^\top \Gamma \theta + \frac{1}{\lambda}\|Y_T - \Phi_T \theta\|_2^2 \right\}$$

follows from completing the square in Gaussian ridge regression. Choosing $\theta = \theta^\star$ cancels the noise quadratic in (9) and proves (10). $\square$

3 MAIR: one identity, two algorithms

3.1 The MAIR objective and gradient bracket

For a dominated model class, a reference belief $\rho \in \Delta(\mathcal{M})$, and an algorithmic belief $\mu \in \Delta(\mathcal{M})$, write

$$\mu_{o|\pi} := \int P_{N,\pi}\, \mu(dN)$$

for the predictive observation law. The model-index algorithmic information ratio objective is

$$\mathsf{MAIR}_{\rho,\eta}(p,\mu) := \mathbb{E}_{M\sim\mu}\Delta_M(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p} I_\mu(M; O \mid \pi) - \frac{1}{\eta}\mathrm{KL}(\mu\|\rho), \tag{11}$$

where

$$I_\mu(M; O \mid \pi) = \mathbb{E}_{M\sim\mu}\, \mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi}).$$

Here, we fix an algorithm and predictive law that sequentially generate

- decision distributions $p_1, \ldots, p_T$, and
- algorithmic beliefs $\mu_1, \ldots, \mu_T$.

The resulting regret bound depends jointly on the decision distributions and the algorithmic belief sequence. We refer to Xu and Zeevi (2025) for the fully measure-theoretic formulation.

Theorem 3.1 (Restatement of Theorem 7.1 in Xu and Zeevi (2025): finite MAIR regret bound)**.** *Consider the stochastic setting with a finite model class $\mathcal{M}$ and a fixed ground-truth model $M^\star \in \mathcal{M}$. Let an arbitrary algorithm produce decision distributions $p_1, \ldots, p_T$, and let $\mu_1, \ldots, \mu_T$ be any sequence of algorithmic beliefs. Let $\rho_t \in \mathrm{int}(\Delta(\mathcal{M}))$ denote the round-t reference belief, generated by the usual posterior update from the previous algorithmic belief, $\rho_{t+1}(\cdot) = \mu_t(\cdot \mid \pi_t, o_t)$. Then, for every $\eta > 0$,*

$$R_T \le \frac{1}{\eta}\mathbb{E}\left[\log \frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)}\right] + \mathbb{E}\left[\sum_{t=1}^{T}\left\{\mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t) + \left\langle \frac{\partial}{\partial\mu}\mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu)\Big|_{\mu=\mu_t}, \mathbf{1}(M^\star) - \mu_t \right\rangle\right\}\right],$$

where $\mathbf{1}(M^\star)$ denotes the vector in $\Delta(\mathcal{M})$ whose $M^\star$ coordinate is one and whose other coordinates are zero. In particular, because $\rho_{T+1}(M^\star) \le 1$, the first term is at most $\log(1/\rho_1(M^\star))/\eta$, and for the uniform prior it is at most $\log|\mathcal{M}|/\eta$.

Lemma 3.2 (MAIR gradient bracket: the unified view)**.** *Assume the map $M \mapsto P_{M,\pi}$ is dominated and the displayed derivatives exist. Fix p, ρ, μ. Up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial\mu(M)}\mathsf{MAIR}_{\rho,\eta}(p,\mu) = \Delta_M(p) - \frac{1}{\eta}\log\frac{\mu(M)}{\rho(M)} - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\,\mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi}).$$

Consequently, for every true model $M^\star$,

$$\begin{aligned}&\mathsf{MAIR}_{\rho,\eta}(p,\mu) + \langle \nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\mu), \delta_{M^\star} - \mu\rangle \\ &\quad = \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\,\mathrm{KL}(P_{M^\star,\pi}\|\mu_{o|\pi}) - \frac{1}{\eta}\log\frac{\mu(M^\star)}{\rho(M^\star)}.\end{aligned} \tag{12}$$

We highlight two different algorithm design principles:

- *when $\mu = \rho$ (e.g., last-round posterior as predictive belief),*

$$\mathsf{MAIR}_{\mu,\eta}(p,\mu) + \langle \nabla_\mu \mathsf{MAIR}_{\mu,\eta}(p,\mu), \delta_{M^\star} - \mu\rangle = \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\,\mathrm{KL}(P_{M^\star,\pi}\|\mu_{o|\pi}). \tag{13}$$

- *when taking $\bar\mu = \arg\max_{\Delta(\mathcal{M})}\mathsf{MAIR}_{\rho,\eta}(p,\mu)$ (maximizing MAIR as predictive belief),*

$$\mathsf{MAIR}_{\rho,\eta}(p,\bar\mu) + \langle \nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\bar\mu),\, \delta_{M^\star} - \bar\mu\rangle \le \mathsf{MAIR}_{\rho,\eta}(p,\bar\mu) \tag{14}$$

Remark 3.3. *The notation $\nabla_\mu \mathsf{MAIR}_{\mu,\eta}(p,\mu)$ means the partial gradient in the second argument of $\mathsf{MAIR}_{\rho,\eta}(p,\mu)$, evaluated at $\rho = \mu$. It does not differentiate the reference belief appearing in the first subscript. The notation $\nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\hat\mu)$ denotes the partial gradient of $\mathsf{MAIR}_{\rho,\eta}(p,\mu)$ with respect to its second argument, differentiated at $\mu = \hat\mu$. Both notations simplify the full Gateaux-derivative notation used in Theorem 3.1; we use these simpler conventions throughout.*

Proof. Write

$$I_\mu(M;O \mid \pi) = \int \mu(dM)\int \log\frac{dP_{M,\pi}}{d\mu_{o|\pi}}\,dP_{M,\pi}.$$

The Gateaux derivative of $\mathbb{E}_\mu\Delta_M(p)$ in the coordinate M is $\Delta_M(p)$. The derivative of $\mathrm{KL}(\mu\|\rho)$ is $\log(\mu(M)/\rho(M)) + 1$, where the $+1$ is a simplex constant. Differentiating the mixture predictive density shows that the derivative of the mutual-information term is $\mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi})$, again up to the same simplex constant. Averaging over $\pi \sim p$ gives the displayed gradient. Pairing this gradient with $\delta_{M^\star} - \mu$ and adding (11) cancels the $\mathbb{E}_\mu\Delta$, mutual-information, and $\mathrm{KL}(\mu\|\rho)$ terms, yielding (12).

We prove the bounds for two choices of μ. In the case $\mu = \rho$, the model-prior log term vanishes, which gives (13). In the case of $\bar\mu \in \arg\max_{\Delta(\mathcal{M})}\mathsf{MAIR}_{\rho,\eta}(p,\mu)$ (MAIR is strongly concave and has unique belief minimizer, see Xu and Zeevi (2025)), the optimality condition gives the (14). □

We conclude that equation (12) is the unifying algebra.

- GP–UCB uses it after choosing the algorithmic belief to be the GP posterior and then verifies the resulting bracket by calibration and optimism. It does not yet bound regret; the regret term must still be controlled; for GP-UCB, that control comes from self-normalized calibration and optimism.
- MAMS uses it before seeing the data by optimizing the bracket over a finite model class. It says that class-wide robust algorithms (e.g., maximizing algorithmic belief) directly lead to regret bound using minimax DEC or maximin MAIR.

3.2 GP–UCB from calibration and optimism

The heterogeneous-prior GP–UCB rule is

$$x_t \in \operatorname*{argmax}_{x\in\mathcal{X}}\left\{\mu^\Gamma_{t-1}(x) + \beta_{t-1}\sigma_{t-1,\Gamma}(x)\right\}. \tag{15}$$

Let $x^\star \in \operatorname{argmax}_x f^\star(x)$. The relevant confidence event is

$$\mathcal{E}_T(\delta) := \left\{\forall t \le T, \forall x \in \mathcal{X} : |f^\star(x) - \mu_{t-1}^\Gamma(x)| \le \beta_{t-1}^\Gamma(\delta)\sigma_{t-1,\Gamma}(x)\right\}. \tag{16}$$

A standard self-normalized argument gives, up to the usual time-uniform logarithmic adjustment,

$$\beta_t^\Gamma(\delta) = \frac{\sqrt{(\theta^\star)^\top \Gamma \theta^\star}}{\sqrt{\lambda}} + \frac{R}{\sqrt{\lambda}}\sqrt{2\left(\mathcal{I}_{t,\Gamma} + \log\frac{1}{\delta}\right)}. \tag{17}$$

The first term is the algorithmic code length of the truth under the heterogeneous prior; it is small for hub truths under the hub-favoring prior and large for cloud truths.

Lemma 3.4 (Self-normalized calibration plus UCB optimism)**.** *On the event* (16)*, heterogeneous-prior GP–UCB satisfies*

$$r_t := f^\star(x^\star) - f^\star(x_t) \le 2\beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x_t).$$

Let $\kappa_\Gamma^2 := \sup_x k_\Gamma(x,x)$ *and assume* $\sigma_{t,\Gamma}^2(x) \le \kappa_\Gamma^2$*. Then*

$$r_t^2 \le 4C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2 g_t^\Gamma, \qquad C_{\lambda,\kappa_\Gamma} := \frac{2\kappa_\Gamma^2}{\log(1+\kappa_\Gamma^2/\lambda)}. \tag{18}$$

Equivalently, for every $\eta > 0$*,*

$$r_t - \frac{1}{\eta} g_t^\Gamma \le \eta C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2. \tag{19}$$

Proof. On (16),

$$f^\star(x^\star) \le \mu_{t-1}^\Gamma(x^\star) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x^\star) \le \mu_{t-1}^\Gamma(x_t) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x_t),$$

where the second inequality is the UCB choice. Another use of the confidence event gives

$$\mu_{t-1}^\Gamma(x_t) \le f^\star(x_t) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x_t).$$

Combining proves the one-step regret bound. For $0 \le u \le \kappa_\Gamma^2$, the function $u/\frac{1}{2}\log(1+u/\lambda)$ is increasing, so

$$u \le C_{\lambda,\kappa_\Gamma}\frac{1}{2}\log(1+u/\lambda).$$

Substitute $u = \sigma_{t-1,\Gamma}^2(x_t)$ and use (6). The offset form (19) is Young's inequality. □

Theorem 3.5 (Heterogeneous GP–UCB as a MAIR algorithm)**.** *On the confidence event* (16)*,*

$$\sum_{t=1}^T r_t \le 2\sqrt{C_{\lambda,\kappa_\Gamma}\mathcal{I}_{T,\Gamma}\sum_{t=1}^T(\beta_{t-1}^\Gamma)^2} \le 2\beta_T^\Gamma\sqrt{C_{\lambda,\kappa_\Gamma} T \mathcal{I}_{T,\Gamma}}. \tag{20}$$

Moreover the MAIR gradient bracket for the posterior reference satisfies the one-step error condition (19)*. Thus*

self-normalized calibration + UCB optimism $\Rightarrow$ *MAIR gradient-error condition.*

Proof. The regret bound follows from (18) and Cauchy–Schwarz:

$$\sum_{t=1}^T r_t \le \sqrt{\left(\sum_{t=1}^T g_t^\Gamma\right)\left(\sum_{t=1}^T \frac{r_t^2}{g_t^\Gamma}\right)} \le 2\sqrt{C_{\lambda,\kappa_\Gamma}\mathcal{I}_{T,\Gamma}\sum_{t=1}^T(\beta_{t-1}^\Gamma)^2}.$$

For the MAIR statement, set $p_t = \delta_{x_t}$ and let the algorithmic belief be the current Gaussian posterior. Lemma 3.2 gives the exact bracket

$$\Delta_{f^\star}(p_t) - \frac{1}{\eta}\mathrm{KL}(P_{f^\star,x_t} \| q_{t,o|x_t}).$$

The KL term is nonnegative, so the bracket is at most r_t. Inequality (19) then gives the advertised MAIR gradient-error certificate with the posterior-reference information increment g_t^Γ. □

This theorem recovers the familiar realized-information proof of GP–UCB. Replacing $\mathcal{I}_{T,\Gamma}$ by the maximal information gain of the heterogeneous kernel k_Γ gives the usual class-wide relaxation. Keeping $\mathcal{I}_{T,\Gamma}$ is the algorithmic version.

3.3 MAMS from the same gradient bracket

Let $\mathcal{M}_\varepsilon$ be a finite discretization of a kernel/RKHS class on a finite action set. For $f \in \mathcal{M}_\varepsilon$ and $x \in \mathcal{X}$, set $r_f(x) = f(x)$ and $P_{f,x} = N(f(x), \lambda)$. For a reference belief $\rho \in \Delta(\mathcal{M}_\varepsilon)$ and an algorithmic belief $\mu \in \Delta(\mathcal{M}_\varepsilon)$, use (11).

Definition 3.6 (Finite kernel MAMS)**.** *Initialize $\rho_1 \in \mathrm{int}(\Delta(\mathcal{M}_\varepsilon))$. At round t, choose a saddle point*

$$(p_t, \mu_t) \in \operatorname*{argmin}_{p \in \Delta(\mathcal{X})} \operatorname*{argmax}_{\mu \in \Delta(\mathcal{M}_\varepsilon)} \mathsf{MAIR}_{\rho_t,\eta}(p, \mu), \tag{21}$$

sample $x_t \sim p_t$, observe Y_t, and update

$$\rho_{t+1}(f) = \frac{\mu_t(f)\varphi_\lambda(Y_t - f(x_t))}{\sum_{g \in \mathcal{M}_\varepsilon} \mu_t(g)\varphi_\lambda(Y_t - g(x_t))}, \tag{22}$$

where φ_λ is the $N(0, \lambda)$ density.

Define the KL-DEC value of the finite class by

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon) := \sup_{\bar{M} \in \mathrm{conv}(\mathcal{M}_\varepsilon)} \inf_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathcal{M}_\varepsilon} \left\{ \Delta_f(p) - \frac{1}{\eta}\mathbb{E}_{x \sim p}\mathrm{KL}(P_{f,x} \| P_{\bar{M},x}) \right\}. \tag{23}$$

Theorem 3.7 (Expected regret of finite kernel MAMS)**.** *If $f^\star \in \mathcal{M}_\varepsilon$, then MAMS satisfies*

$$\mathfrak{R}_T(\mathsf{MAMS}, f^\star) \le \frac{\log(1/\rho_1(f^\star))}{\eta} + \mathbb{E}\left[\sum_{t=1}^T \mathsf{MAIR}_{\rho_t,\eta}(p_t, \mu_t)\right] \le \frac{\log(1/\rho_1(f^\star))}{\eta} + T\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon). \tag{24}$$

For the uniform prior, the first term is $\log|\mathcal{M}_\varepsilon|/\eta$.

This shows that the regret bound of MAMS is always no worse than the regret bound of the Estimation-to-Decision (E2D) algorithm in Foster et al. (2021).

Proof. Apply Theorem 3.1. With a nonuniform prior the first term is $\log(1/\rho_1(f^\star))/\eta$. The saddle-point choice makes μ_t a maximizer of $\mathsf{MAIR}_{\rho_t,\eta}(p_t, \mu)$, so the first-order condition gives

$$\langle \nabla_\mu \mathsf{MAIR}_{\rho_t,\eta}(p_t, \mu_t), \delta_{f^\star} - \mu_t \rangle \le 0.$$

Therefore the regret is bounded by the prior code length plus $\mathbb{E}\sum_t \mathsf{MAIR}_{\rho_t,\eta}(p_t, \mu_t)$. By the minimax choice of p_t and the MAIR-to-DEC comparison, each summand is at most $\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon)$. Summing proves the claim. □

Corollary 3.8 (Uniform prior and discretization bookkeeping). *Suppose $\mathcal{M}_\varepsilon$ is an ε-net of a larger kernel class in sup norm and every $f^\star$ has a projection $\hat{f} \in \mathcal{M}_\varepsilon$ with $\|f^\star - \hat{f}\|_\infty \le \varepsilon$. If, in addition, the observation channels satisfy*

$$\sup_x \mathrm{KL}(P_{f^\star,x} \| P_{\hat{f},x}) \le \chi_\varepsilon,$$

then the same proof with a misspecification comparison yields the schematic bound

$$\mathfrak{R}_T(\mathsf{MAMS}, f^\star) \le \frac{\log|\mathcal{M}_\varepsilon|}{\eta} + T\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon) + 2\varepsilon T + \frac{\chi_\varepsilon T}{\eta}. \tag{25}$$

For Gaussian observations with common variance λ, one may take $\chi_\varepsilon = \varepsilon^2/(2\lambda)$.

Proof. The regret gap under $f^\star$ and under $\hat{f}$ differs by at most 2ε per round. The likelihood-ratio part of the MAIR proof changes by at most χ_ε/η per round. Applying Theorem 3.7 to $\hat{f}$ and adding these two comparison terms gives (25). The Gaussian value of χ_ε is the KL divergence between two one-dimensional Gaussians with equal variance and means separated by at most ε. □

This is the precise sense in which MAMS provides a finite-covering route to kernel bandits. It is robust and DEC-aligned, but the covering term $\log|\mathcal{M}_\varepsilon|$ and the misspecification bookkeeping can be much larger than the log-determinant information accounting used by GP–UCB. This is why GP–UCB is often the practical algorithm, even if MAMS is the clean class-wide minimax comparator.

4 Comparing GP–UCB, MAMS, and minimax benchmarks

4.1 What it means to compare the algorithms

GP–UCB and MAMS should be compared on the same DMSO instance, but not by collapsing everything to one scalar too early.

	Heterogeneous GP–UCB	**MAMS / DEC**
Decision rule	Deterministic optimistic acquisition $\mu + \beta\sigma$ under an algorithmic covariance.	Randomized saddle point of $\inf_p \sup_\mu \mathsf{MAIR}_{\rho,\eta}(p, \mu)$.
Belief/reference	Fixed posterior path induced by the chosen prior geometry Γ.	Optimized adversarial belief and posterior reference.
Information term	Realized log determinant $\mathcal{I}_{T,\Gamma}$.	Prior code length plus DEC/MAIR envelope.
Strength	Practical, kernel-algebraic, often tight on estimation/information gain.	Robust, class-wide, designed to match minimax lower-bound language.
Weakness	Coefficient controlled by confidence widths, rather than optimized as in DEC; a biased prior may fail on high-complexity truths.	Finite-covering implementation may overpay estimation/discretization cost.

The comparison that seems most informative experimentally is therefore three-dimensional: fixed-truth expected regret (1), Bayesian-average regret under a chosen reference prior, and class-wide minimax or DEC value.

4.2 A safeguarded route: no worse than MAMS, good on structured truths

A single heterogeneous prior cannot have everything for free. If the prior assigns tiny variance to cloud directions, GP–UCB will be excellent on hub truths but may be slow on cloud truths. If the prior assigns large variance to every direction, it is robust but loses the hub-scale advantage. A clean way to express the desired guarantee is to safeguard the practical algorithm with a robust base algorithm.

Proposition 4.1 (Meta-oracle form of a safeguarded algorithm)**.** *Suppose a bandit-over-bandit master combines two base algorithms, heterogeneous GP–UCB and finite-kernel MAMS, and guarantees for every model M an overhead $\mathfrak{C}_T$ relative to the better base algorithm:*

$$\mathfrak{R}_T(\mathsf{Safe}, M) \leq \min\{\mathfrak{R}_T(\mathsf{UCB}_\Gamma, M), \mathfrak{R}_T(\mathsf{MAMS}, M)\} + \mathfrak{C}_T.$$

Then

$$\mathfrak{R}_T(\mathsf{Safe}, M) \leq \mathfrak{R}_T(\mathsf{MAMS}, M) + \mathfrak{C}_T \qquad \text{for all } M \in \mathcal{M},$$

while on any structured subfamily $\mathcal{M}_H$ on which heterogeneous GP–UCB has an algorithmic bound $B_H(T)$,

$$\sup_{M \in \mathcal{M}_H} \mathfrak{R}_T(\mathsf{Safe}, M) \leq B_H(T) + \mathfrak{C}_T.$$

Proof. Both claims follow by taking the corresponding term inside the minimum. Standard corralling algorithms provide such regret-to-best-base guarantees for bandit algorithms under the usual feedback assumptions (Agarwal et al., 2017); the present proposition only records the consequence needed for the comparison. □

This proposition gives the right conceptual target. Heterogeneous GP–UCB explains the favorable algorithmic behavior. MAMS supplies the class-wide insurance. When the finite MAMS component is tuned to the DEC/minimax rate of the chosen cover, the safeguarded master is not worse than that class-wide benchmark up to the cover error and the master overhead, while still inheriting the hub-scale GP–UCB bound on structured truths. The main theorems of this paper do not rely on the master; it is included to clarify why the algorithmic-prior view is compatible with, rather than opposed to, minimax robustness.

5 Hub–cloud separation and experimental protocol

5.1 A heterogeneous-prior hub–cloud construction

The hub–cloud perspective is based on the pointwise-complexity separation of Xu (2026). Let $\mathcal{X} = H_M \cup C_N$, where $H_M = \{h_0, h_1, \ldots, h_M\}$ is a hub/leaf set and $C_N = \{c_1, \ldots, c_N\}$ is a large cloud. Use independent-arm features, so the ambient kernel is diagonal. Choose a heterogeneous algorithmic covariance

$$k_\Gamma(h, h) = 1 \quad (h \in H_M), \qquad k_\Gamma(c, c) = \varepsilon_C^2 \quad (c \in C_N),$$

with $\varepsilon_C \ll 1$. Equivalently, the prior precision is much larger on cloud coordinates. The GP–UCB acquisition is the ordinary one for this heterogeneous kernel; no additional acquisition penalty is needed.

Consider hub truths $\mathcal{M}_H$ in which one hub arm $h^\star$ has mean $R > 0$, the other hub arms have mean 0, and all cloud arms have mean 0. These truths have small norm in the heterogeneous RKHS because their cloud coordinates vanish. Consider also cloud truths $\mathcal{M}_C$ in which one cloud arm has mean $S > 0$ and all other arms have mean 0. These have norm of order S/ε_C and are high-complexity under the hub-favoring prior.

Theorem 5.1 (Cloud minimax lower bound and DEC certificate)**.** *Let M_j, $j \in [N]$, denote the cloud model in which arm c_j has mean reward $S > 0$ and every other arm has mean reward zero. Pulling arm x under M_j produces an observation distributed as*

$$Y \mid x, M_j \sim N(r_{M_j}(x), \lambda), \qquad \lambda > 0,$$

independently across rounds conditional on the chosen arms. Let $\mathcal{M}_C = \{M_1, \ldots, M_N\}$. Then every adaptive bandit algorithm satisfies, for $T \le N/4$,

$$\sup_{M \in \mathcal{M}_C} \mathfrak{R}_T(\mathsf{ALG}, M) \ge \frac{3}{4} ST. \tag{26}$$

Consequently, since the display holds for every algorithm,

$$\mathfrak{R}_T^{\mathrm{mm}}(\mathcal{M}_H \cup \mathcal{M}_C) \ge \frac{3}{4} ST.$$

For the DEC side, let M_0 be the null model with zero mean reward and zero observation mean on every arm, and set $\mathcal{M}^0_{\mathrm{amb}} := \mathcal{M}_H \cup \mathcal{M}_C \cup \{M_0\}$. With the KL-DEC convention in (23),

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}^0_{\mathrm{amb}}) \ge S\left(1 - \frac{1}{N}\right) - \frac{S^2}{2\eta\lambda N} \qquad \textit{for every } \eta > 0. \tag{27}$$

Equivalently, the reference-specific DEC offset against the null reference M_0 already has the lower bound on the right side of (27). If one insists on the unaugmented class and uses only convex references generated by the cloud models themselves, then the weaker but still cloud-sensitive bound

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_H \cup \mathcal{M}_C) \ge S\left(1 - \frac{1}{N}\right) - \frac{\log N + 2}{\eta N} \tag{28}$$

holds for $N \ge 2$. Thus the class-wide DEC certificate sees the cloud unless the reference or model class is localized away from $\mathcal{M}_C$.

Proof. We first prove the minimax lower bound. Put the uniform prior on the cloud index $J \sim \mathrm{Unif}([N])$ and run the algorithm on the corresponding model M_J. Let

$$\tau := \inf\{t \ge 1 : x_t = c_J\}$$

be the first time the algorithm samples the good cloud arm, with the convention $\tau = \infty$ if this never occurs. Couple this experiment with the null experiment M_0 as follows. Under M_0 all observations are $N(0, \lambda)$. Draw $J \sim \mathrm{Unif}([N])$ independently of the null observations and let $A_T^0 \subseteq [N]$ be the set of cloud indices sampled by the algorithm during the first T rounds of the null experiment. Since $|A_T^0| \le T$ almost surely,

$$\mathbb{P}_0\{J \in A_T^0\} = \mathbb{E}_0\left[\frac{|A_T^0|}{N}\right] \le \frac{T}{N}.$$

Before the arm c_J is sampled, the observations under M_J have exactly the same distribution as under the null model along the realized action path. Hence the event that the true experiment hits c_J by time T is contained, under the above coupling, in the event $\{J \in A_T^0\}$, and therefore

$$\mathbb{P}\{\tau \le T\} \le \frac{T}{N} \le \frac{1}{4}.$$

On the event $\{\tau > T\}$ the algorithm never plays the unique optimal arm during the first T rounds, so it suffers regret S in every round. Thus the Bayes regret under the uniform cloud prior is at least

$$\mathbb{E}_J \mathfrak{R}_T(\mathsf{ALG}, M_J) \ge ST\, \mathbb{P}\{\tau > T\} \ge \frac{3}{4} ST.$$

Since a supremum is at least an average, (26) follows.

We next prove the DEC lower bound. Because $M_0 \in \mathcal{M}^0_{\mathrm{amb}} \subseteq \mathrm{conv}(\mathcal{M}^0_{\mathrm{amb}})$, the robust KL-DEC is at least the value obtained by fixing the convex reference to be M_0:

$$\mathsf{DEC}^{\mathrm{KL}}_\eta(\mathcal{M}^0_{\mathrm{amb}}) \geq \inf_{p \in \Delta(\mathcal{X})} \sup_{j \in [N]} \left\{ \Delta_{M_j}(p) - \frac{1}{\eta} \mathbb{E}_{x \sim p} \mathrm{KL}(P_{M_j,x} \| P_{M_0,x}) \right\}.$$

Fix any decision distribution $p \in \Delta(\mathcal{X})$ and write $p_j := p(c_j)$. Since $\sum_{j=1}^N p_j \leq 1$, there exists $j_p \in [N]$ with $p_{j_p} \leq 1/N$. Under model M_{j_p} the optimal arm is c_{j_p} and has reward S, while every other arm has reward zero. Therefore

$$\Delta_{M_{j_p}}(p) = S(1 - p_{j_p}) \geq S\left(1 - \frac{1}{N}\right).$$

Moreover, the only arm whose observation law differs between M_{j_p} and M_0 is c_{j_p}; for this arm the two laws are $N(S, \lambda)$ and $N(0, \lambda)$. Hence

$$\mathbb{E}_{x \sim p} \mathrm{KL}(P_{M_{j_p},x} \| P_{M_0,x}) = p_{j_p} \frac{S^2}{2\lambda} \leq \frac{S^2}{2\lambda N}.$$

Combining the two displays gives, for this j_p,

$$\Delta_{M_{j_p}}(p) - \frac{1}{\eta} \mathbb{E}_{x \sim p} \mathrm{KL}(P_{M_{j_p},x} \| P_{M_0,x}) \geq S\left(1 - \frac{1}{N}\right) - \frac{S^2}{2\eta\lambda N}.$$

Since the argument holds for every p, taking $\sup_j$ and then $\inf_p$ proves (27).

It remains to prove the unaugmented convex-reference variant. Let $\bar{M}_C$ be the uniform mixture of the cloud models. This is an admissible reference in $\mathrm{conv}(\mathcal{M}_H \cup \mathcal{M}_C)$. Fix again $p \in \Delta(\mathcal{X})$ and choose j_p with $p_{j_p} \leq 1/N$. Under M_{j_p} the regret gap is still at least $S(1 - 1/N)$. We bound the KL term against the mixture reference. At arm c_{j_p}, the mixture observation density contains the true density with weight $1/N$, so

$$\mathrm{KL}(P_{M_{j_p},c_{j_p}} \| P_{\bar{M}_C,c_{j_p}}) \leq \log N.$$

At a different cloud arm c_ℓ, $\ell \neq j_p$, the law under M_{j_p} is the null Gaussian density, while the mixture density contains this null Gaussian component with weight $1 - 1/N$. Hence

$$\mathrm{KL}(P_{M_{j_p},c_\ell} \| P_{\bar{M}_C,c_\ell}) \leq -\log\left(1 - \frac{1}{N}\right) \leq \frac{2}{N},$$

where the last inequality uses $N \geq 2$. Hub arms have zero mean under both M_{j_p} and $\bar{M}_C$, and therefore contribute zero KL. Thus

$$\mathbb{E}_{x \sim p} \mathrm{KL}(P_{M_{j_p},x} \| P_{\bar{M}_C,x}) \leq p_{j_p} \log N + \frac{2}{N} \sum_{\ell \neq j_p} p(c_\ell) \leq \frac{\log N + 2}{N}.$$

Using the reference $\bar{M}_C$ in the outer supremum of the KL-DEC definition and repeating the previous offset argument gives (28). □

The separation is expressed entirely through the GP–UCB prior geometry. The same algorithmic covariance that gives the upper bound suppresses the cloud in the construction. This also reveals the limitation: the algorithm is not designed to be good on cloud truths. Proposition 4.1 describes how to add a MAMS safeguard if one wants both class-wide insurance and hub-scale adaptation.

5.2 Experimental protocol: algorithmic, Bayesian, and minimax curves

The experiments should compare three quantities, not only regret.

Bayesian GP sample paths. Draw $f \sim \mathsf{GP}(0, k)$ on a finite grid for Matérn and squared-exponential kernels. Run GP–UCB, MVR, finite-grid MAMS, and the safeguarded master. Plot cumulative regret and realized information traces $\mathcal{I}_{T,\Gamma}$. The simulations of Iwazaki (2025) already show that GP–UCB can collect much less realized information than MVR; adding MAMS shows whether the robust MAIR/DEC rule behaves more like variance reduction or more like optimism.

Frequentist RKHS functions. Choose deterministic $f^\star$ with controlled heterogeneous norm: a smooth unimodal function, a multi-peak function, hub truths, and cloud truths. Report fixed-truth expected regret (1), realized information, and local ratios r_t^2/g_t. This directly tests the algorithmic-information certificate.

Class-wide stress test. On finite grids, approximate $\sup_{f\in\mathcal{M}_\varepsilon} \mathfrak{R}_T(\mathsf{ALG}, f)$ and compare it with the finite MAMS guarantee and lower-bound estimates. The expected outcome is not that one algorithm dominates all criteria. GP–UCB should often be best on favorable structured truths and have small realized information. MAMS should better track the robust DEC envelope. The safeguarded master should interpolate between the two.

6 Relation to DEC, Bellman rank, and the GP–UCB optimality debate

6.1 DEC, Bellman rank, and decision/model duality

This paper uses DMSO in the standard sense: the decision space, model class, observation laws, and rewards are explicitly specified. DEC is powerful in this setting because it is a clean class-wide offset complexity. A phrase such as "bounded Bellman rank" is different. Bellman rank is a representation-relative algebraic certificate depending on value-function classes, roll-in distributions, and Bellman-error witnesses (Jiang et al., 2017). Without fixing the witness, discrepancy, and observable experiment, bounded Bellman rank is not itself a DMSO model class and cannot be assigned an intrinsic DEC.

Bellman rank says that there exists a low-dimensional witness structure under which a class is learnable. DEC says that a fixed observable model class has a particular decision-estimation tradeoff. Once a bilinear-class witness is fixed and its estimation complexity is charged, one can ask DEC-style questions (Du et al., 2021). Without that, a rich witness class can make a lower bound non-intrinsic or vacuous.

There are also gaps within clean DMSO classes. Chen et al. (2024) show that DEC-style lower bounds may need estimation-complexity refinements such as fractional covering to recover tight lower bounds in general interactive problems. Work on ridge-bandit and even MAB models shows another lesson: minimax regret may not be recovered from a scalar one-round coefficient alone; algorithm dynamics and phases matter (Rajaraman et al., 2023; Gu et al., 2025).

The hub–cloud separation here is different. It does not exploit representation ambiguity or an omitted estimation term. The class is finite, the observations are Gaussian, and the regret loss is ordinary. The full-class minimax/DEC benchmark is large because the class contains a huge cloud of high-complexity alternatives. A heterogeneous-prior GP–UCB rule intentionally refuses to be robust to those alternatives unless evidence accumulates. Its algorithmic AIR performance on hub truths is therefore much smaller than the full-class certificate.

Ignoring historical influence and focusing on mathematical form, IR/AIR and DEC are dual offset views. IR/MAIR conditions on the algorithmic belief and decision/model marginal, then measures how much posterior-update information the algorithm obtains from its own action. DEC fixes a model class and reference model, then asks for the best decision distribution against the worst model. IR/AIR is naturally suited to concrete algorithms: TS, IDS, exponential weights, inverse-propensity rules, and, as shown here, UCB families after calibration. DEC is naturally suited to minimax lower bounds, robust class-wide algorithms, and black-box estimators.

	Class-wide coefficient	**Algorithmic coefficient**
IR / AIR / MAIR	Often summarized by a scalar dimension or rank; useful for upper bounds once calibrated.	Realized information plus local regret-information coefficient; applies to actual policies such as GP–UCB.
DEC / convex-hull variants	Intrinsic to a specified model class; excellent for minimax lower bounds and robust algorithms.	Localized DEC variants can certify subfamilies, but the localization must be part of the algorithm/model specification.

6.2 What the GP–UCB optimality debate becomes in MAIR language

The classical GP–UCB proof has two factors: cumulative information and the coefficient converting confidence width into regret. The posterior-reference theorem sharpens the first factor by keeping the realized log determinant $\mathcal{I}_{T,\Gamma}$ instead of immediately replacing it by γ_T. The hard remaining issue is the coefficient. In the standard analysis, calibration gives β_t^2 of order the information scale, so the regret bound behaves like an information factor times $\sqrt{T}$.

Recent progress should be read through this lens. Whitehouse et al. (2023) improve GP–UCB rates by changing the calibration analysis, using smoothness-aware regularization and sharper Hilbert-space concentration. In Bayesian GP bandits, Iwazaki (2025) show that the realized GP–UCB trajectory has much smaller information behavior than worst-case MVR-style designs in important kernels. These are not contradictions: one improves the frequentist calibration coefficient, the other improves realized information accounting. MAIR separates the two.

This is also where MAMS/E2D is informative but not a drop-in replacement. MAMS/E2D optimizes the offset coefficient in a finite class and therefore fits the DEC lower-bound language. But passing MAMS/E2D through a kernel covering can overpay finite estimation complexity. GP–UCB, by contrast, uses kernel linear algebra and log determinants directly, but its UCB rule is not designed to minimize the MAIR coefficient. The most promising practical direction is a coefficient-aware kernel algorithm that keeps GP–UCB's posterior computations while borrowing the MAMS/E2D lesson about robust offset optimization.

6.3 Conclusion

The paper's view is deliberately pluralistic. Class-wide minimax and DEC lower bounds remain the correct tools for robust guarantees over a specified ambient class. GP–UCB remains the practical kernel algorithm because it uses a computationally tractable posterior geometry and often has favorable realized information. The algorithmic-information language explains both facts at once.

Changing scalar ridge regularization to a heterogeneous PSD prior is not a cosmetic generalization. It exposes the algorithm's inductive bias as a mathematical object: the same matrix controls the posterior variance, the information gain, the fixed-truth code length, and the hub–cloud separation. A biased prior can be excellent for structured truths and poor in the worst case. A safeguarded GP–UCB/MAMS master can recover worst-case insurance while preserving the algorithmic gain. This is the sense in which algorithmic performance, rather than class-wide minimax risk alone, should be part of the kernel/GP bandit optimality discussion.

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